

```

10
11 const int DataPin = 35; //19;
12 const int IRQpin = 34; //18;
13 PS2Keyboard keyboard;
14
15 byte x, y; // track lcd position
16 int datalength;
17
18 int a=0;
19 int b=5;
20 String msg="";
21 const char backslash = 3;
22 char c;
23
24 const char* ssid = "beral"; //use your own id
25 const char* password = "*****"; //use your own password
26 const char* api_key = "*****"; //create your own api key, see detail in write up
27
28 void connectToWiFi() {
29   WiFi.begin(ssid, password);
30   while (WiFi.status() != WL_CONNECTED) {
31     delay(1000);

```

Fig. 5: Snippet of source code configuration of the Wi-Fi SSID and password

```

56
57 String makeApiRequest(String prompt1) {
58   HTTPClient http;
59   http.setTimeout(24000); // 24000 Set a longer timeout (in milliseconds)
60   http.begin("https://api.openai.com/v1/chat/completions");
61   http.addHeader("Content-Type", "application/json");
62   http.addHeader("Authorization", "Bearer " + String(api_key));
63
64   StaticJsonDocument<1024> jsonPayload;
65   jsonPayload["model"] = "gpt-3.5-turbo";
66   jsonPayload["temperature"] = 0.8;
67   jsonPayload["max_tokens"] = 2000;
68
69   JsonArray messages = jsonPayload.createNestedArray("messages");
70   JsonObject userMessage = messages.createNestedObject();
71   userMessage["role"] = "user";
72   userMessage["content"] = prompt1;
73
74   String payloadString;
75   serializeJson(jsonPayload, payloadString);
76
77   int httpStatusCode = http.POST(payloadString);
78   if (httpStatusCode == HTTP_CODE_OK) {

```

Fig. 6: Snippet of the source code configured to the OpenAI model

ESP32. Refer to Table 2 and Table 3, and Fig. 2 and Fig. 3, for the connections.

For connecting PS2 keyboard, refer to Fig. 3 and Table 3. A PS2 keyboard, like a USB keyboard, needs only 4 pins to connect to an MCU: 5V, ground, data pin, and clock pin (or IRQ pin). Since this is a pure data-in device, GPIO pins 34 and 35 can be connected to it. GPIO pins 34 and 35 are data-in pins only; they are not data-out pins. Never use them for TFT, LEDs, relays, etc as they require signal output, unlike keyboard pins that need data input. Of course, you can connect a PS2 keyboard to other GPIO pins as well.

Here, the ESP is used to connect to the internet and ChatGPT using the OpenAI API, and the TFT display is used as terminal display for the ChatGPT output. The query can be entered using the PS2 keyboard.

You may wonder why an old PS2 keyboard is being used in the prototype instead of a USB keyboard. I tried to connect a sleek, small USB keyboard, but I could not get it to work with the ESP32.

Software coding

First, you need to create an OpenAI

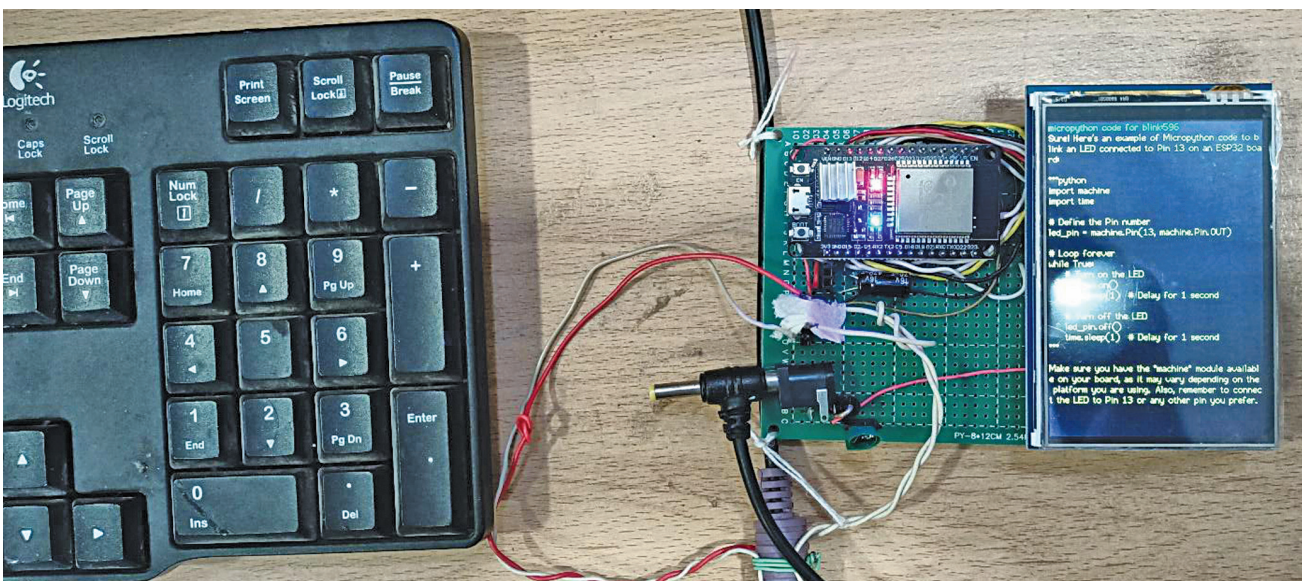


Fig. 7: Final assembled terminal used for testing